Design and Implement a Basic URL Shortener Using Spring Boot

Date: 22.07.2026 Name: Amrit Gopinath
Ex. No: 2 Reg No: 3122 24 5001 015

Aim: To design and implement a basic URL Shortener application using Spring Boot that generates a short URL for a given long URL using a hashing technique, and redirects users to the original URL.

Software Used and its Configuration

Software	Role	Purpose in this Assignment
Spring Boot 4.1.0	Application Framework	Runs the Java application; routes HTTP requests to UrlController, wires up the service and repository layers, starts an embedded web server.
MongoDB	Database	Stores URL collection (id, shortURL, longURL), connected at mongodb://localhost:27017/urlshortner
Postman	API Client	Used to call POST /url/shorten and GET /url/{shortCode} manually while testing the endpoints.
VS Code	Editor / IDE	Where the API is built.

Spring Boot Configuration

Spring Boot itself is configured, not installed separately, through the dependencies (<dependency> entries) declared in **pom.xml**.

1. spring-boot-starter-webmvc
    - REST controller support.
2. spring-boot-starter-data-mongodb
    - The MongoDB driver and MongoRepository support.

Lombok

Lombok generates and writes getters, setters, and constructors for Entities and DTOs.

application.properties Configuration

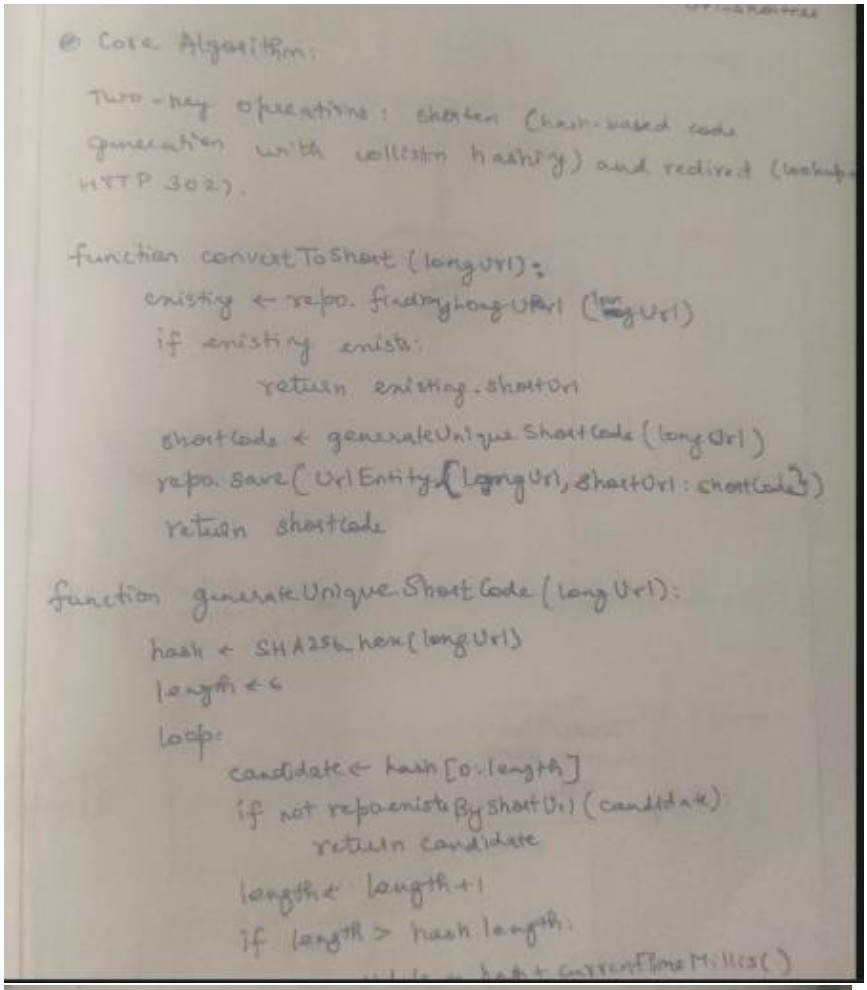
1. spring.application.name = url-shortner
2. spring.mongodb.uri = mongodb://localhost:27017/url-shortner

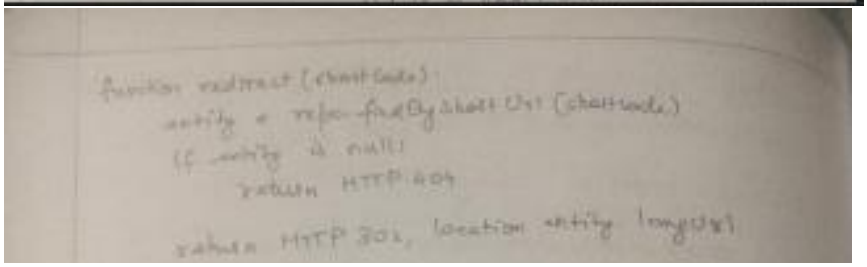
Design and Implement a Basic URL Shortener Using Spring Boot

Date: 22.07.2026
Ex. No: 2

Name: Amrit Gopinath
Reg No: 3122 24 5001 015

Core Algorithm:



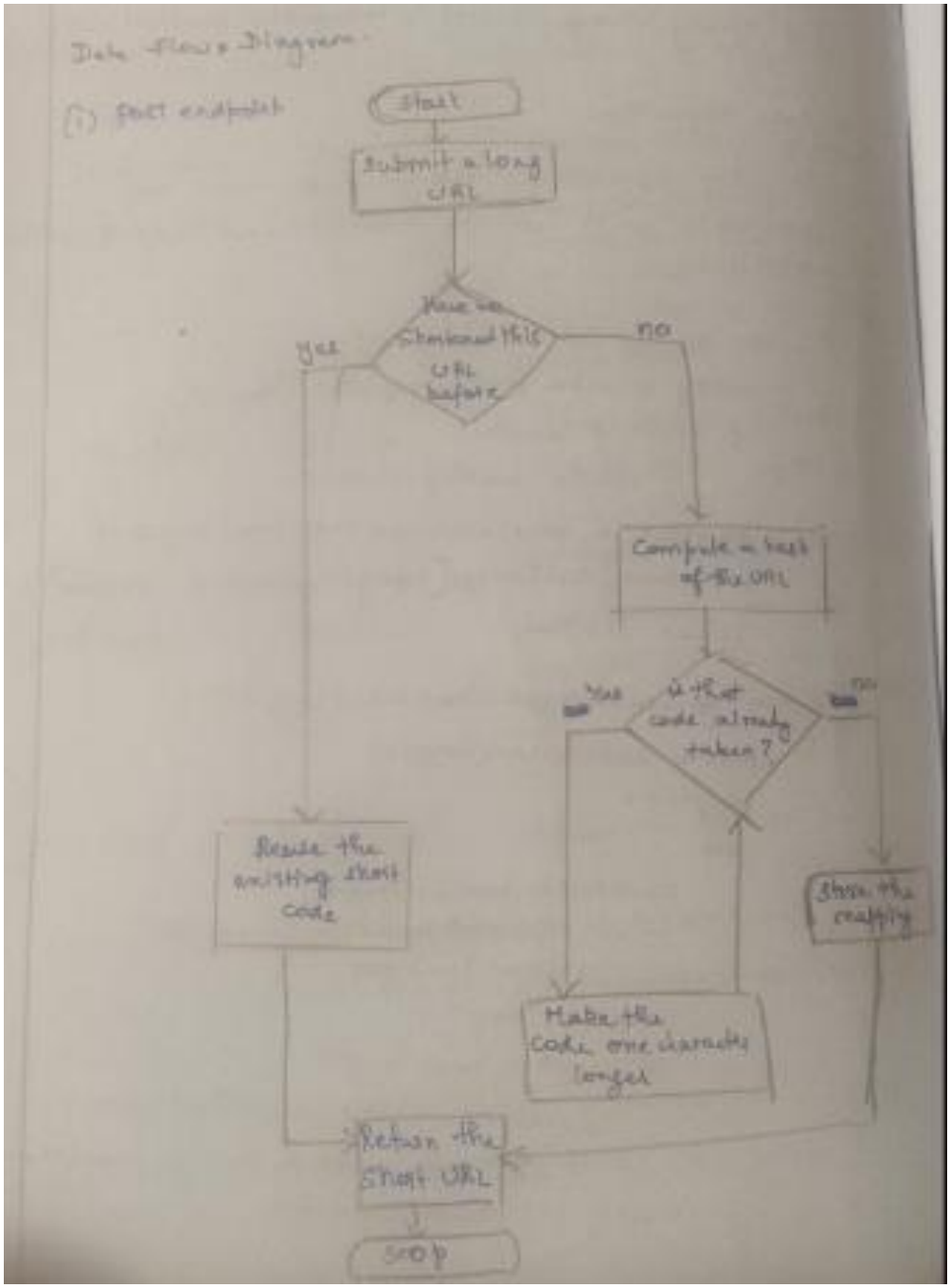


Design and Implement a Basic URL Shortener Using Spring Boot

Date: 22.07.2026
Ex. No: 2

Name: Amrit Gopinath
Reg No: 3122 24 5001 015

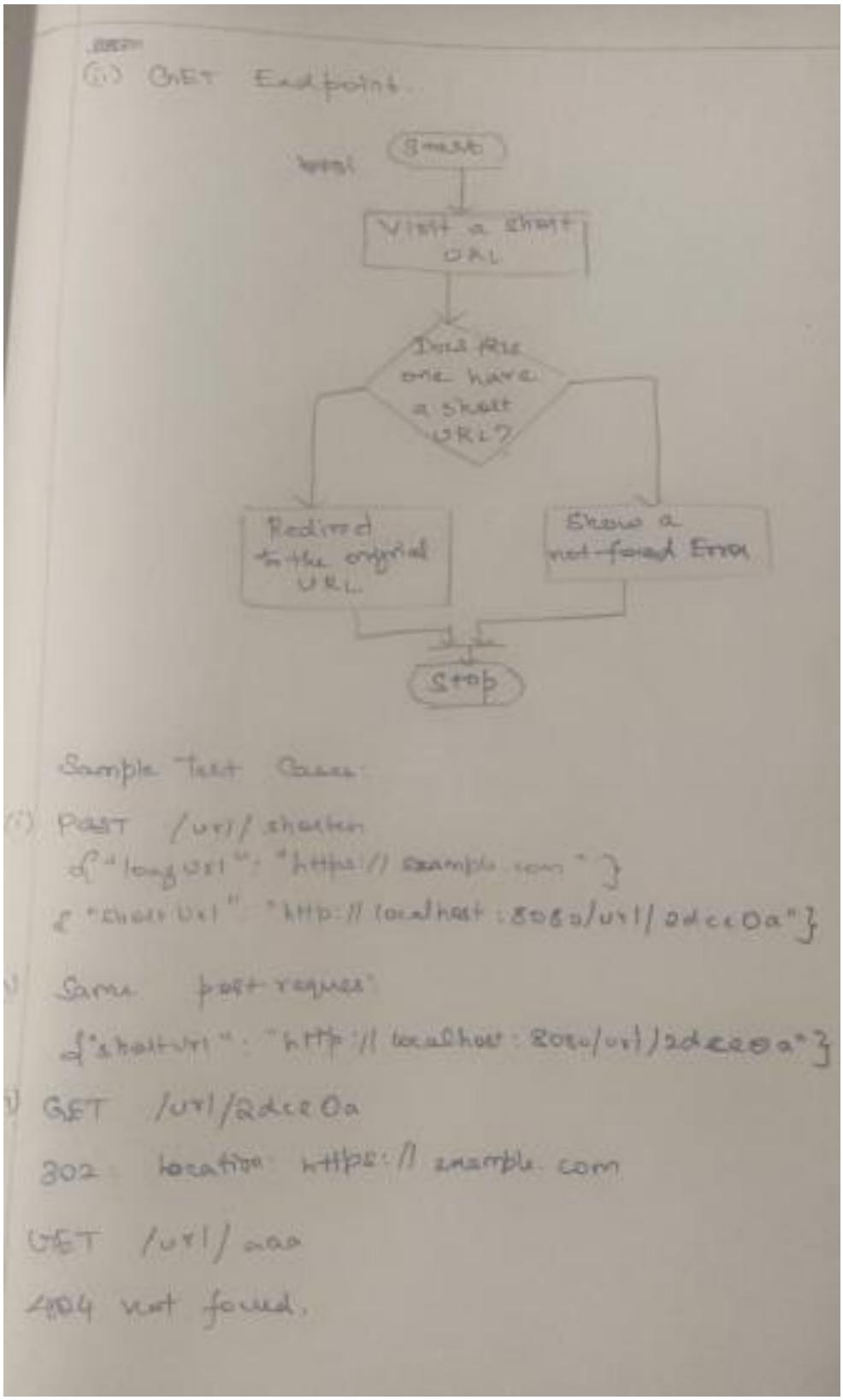
System Design and Flow:



Design and Implement a Basic URL Shortener Using Spring Boot

Date: 22.07.2026
Ex. No: 2

Name: Amrit Gopinath
Reg No: 3122 24 5001 015

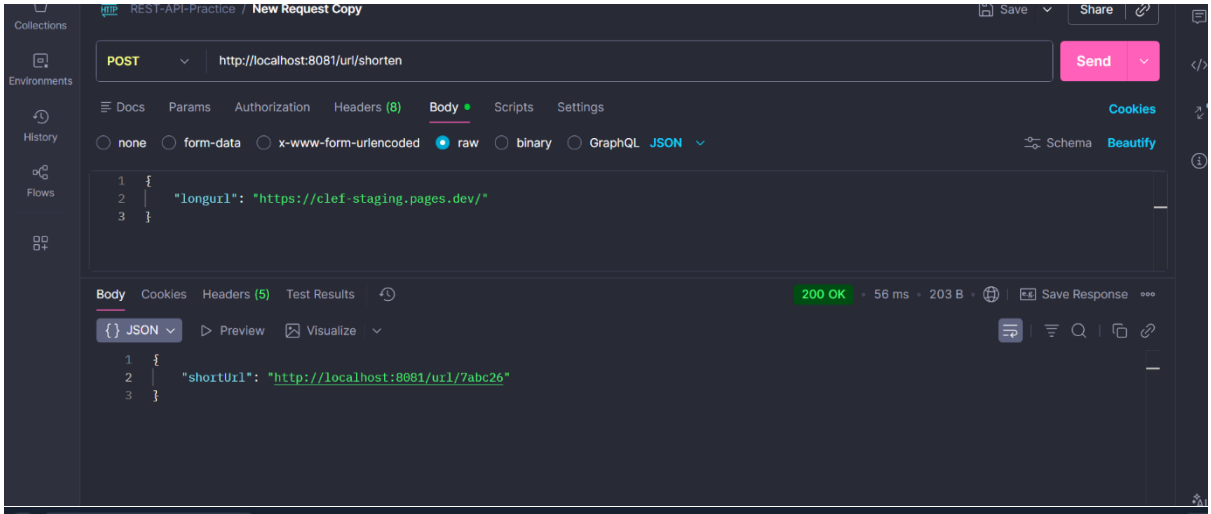


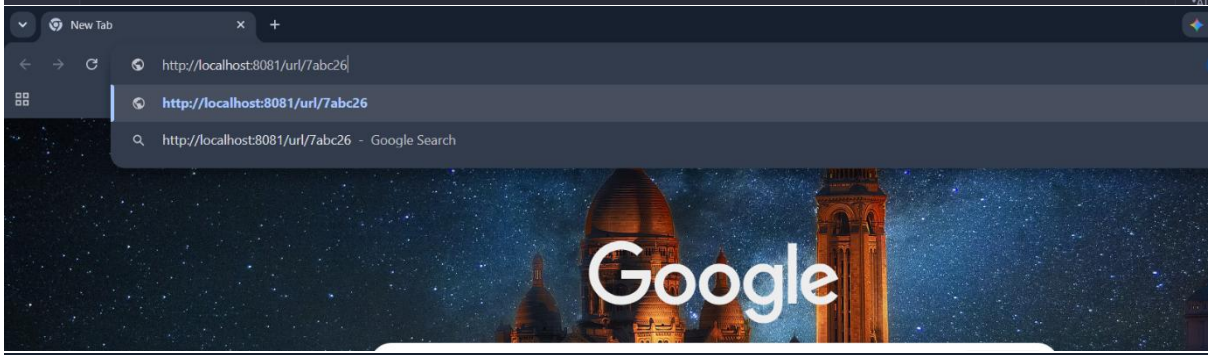
Design and Implement a Basic URL Shortener Using Spring Boot

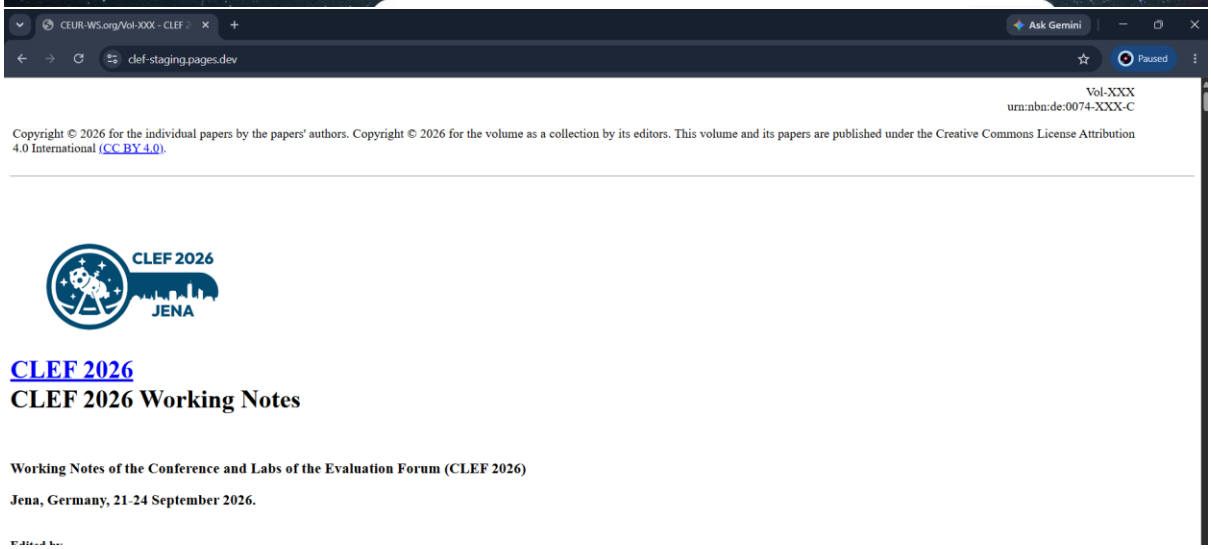
Date: 22.07.2026
Ex. No: 2

Name: Amrit Gopinath
Reg No: 3122 24 5001 015

Test Cases (Screenshots):





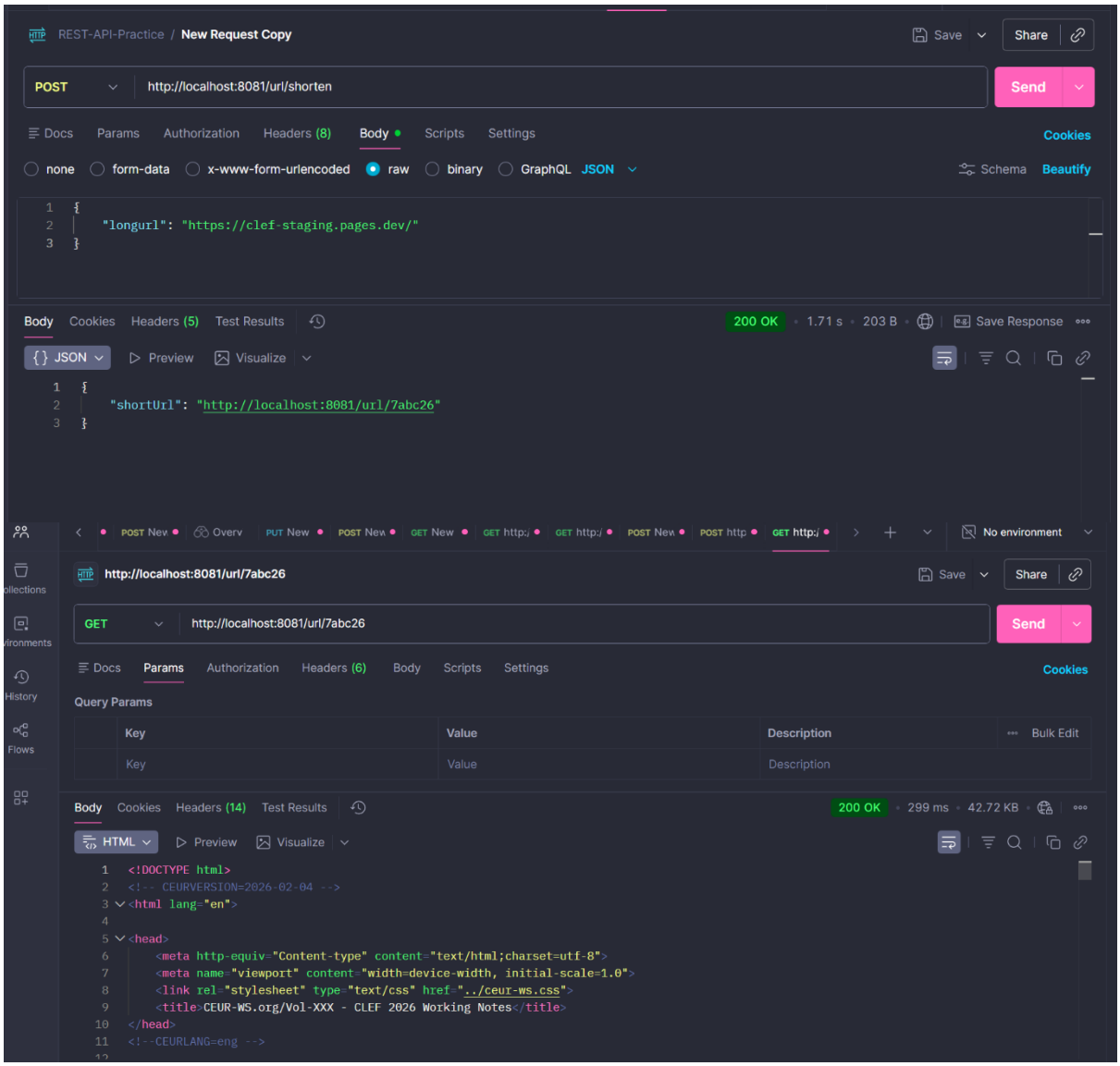


Design and Implement a Basic URL Shortener Using Spring Boot

Date: 22.07.2026
Ex. No: 2

Name: Amrit Gopinath
Reg No: 3122 24 5001 015

Resending the same URL in the request:

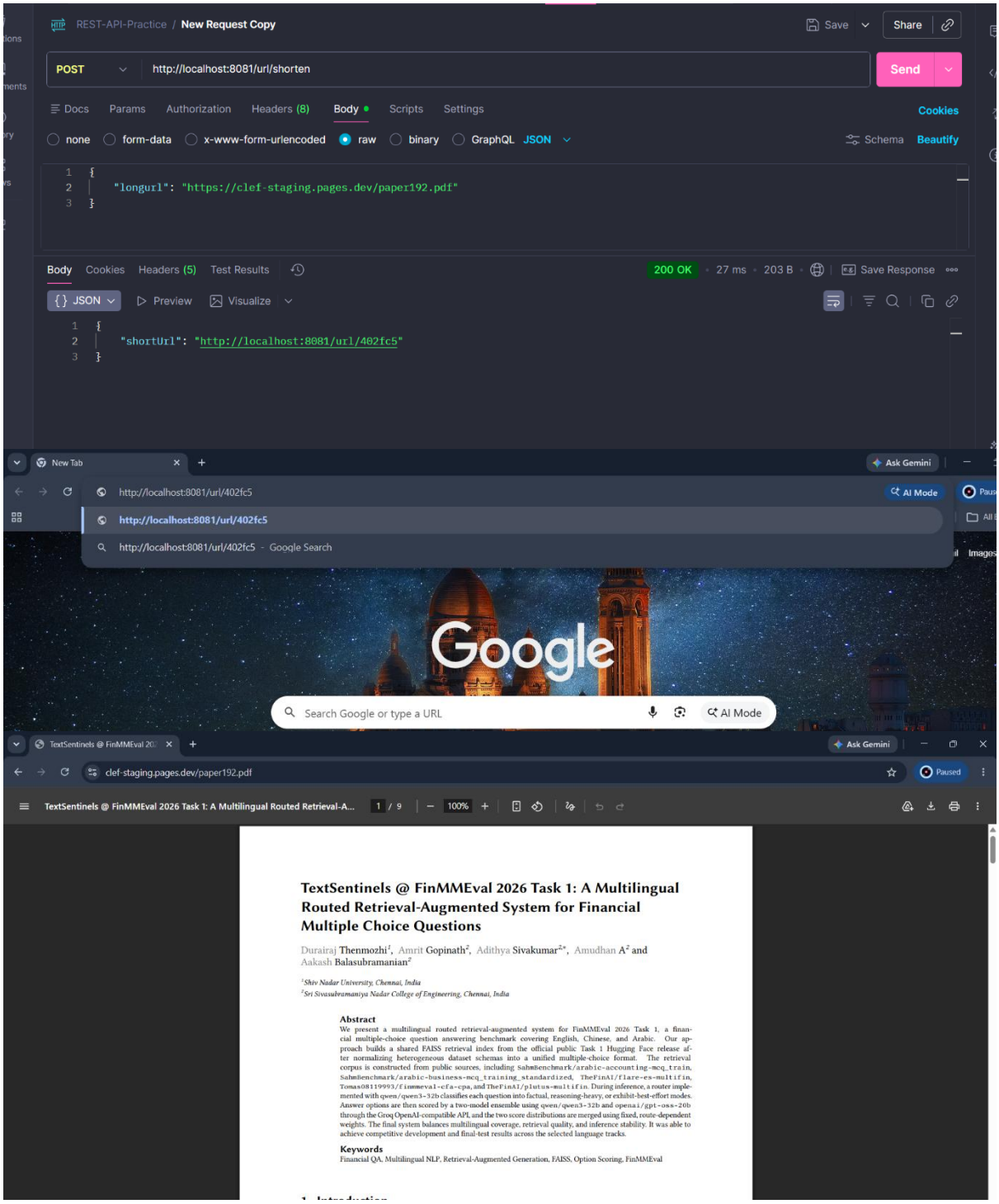


Design and Implement a Basic URL Shortener Using Spring Boot

Date: 22.07.2026
Ex. No: 2

Name: Amrit Gopinath
Reg No: 3122 24 5001 015

Test Case 2:



Design and Implement a Basic URL Shortener Using Spring Boot

Date: 22.07.2026 Name: Amrit Gopinath
Ex. No: 2 Reg No: 3122 24 5001 015

Learning Outcomes:

- Design and develop a URL Shortener application using Spring Boot and a layered architecture.
- Implement RESTful APIs to generate unique short URLs using hashing techniques and perform URL redirection.
- Store and retrieve URL mappings efficiently using MongoDB and Spring Data MongoDB.